Geothermal Energy

Chapter 8 Unit 2 Guided Notes · Grade 4 Using Natural Resources for Energy

Name: _____ **Date:** _____

BIG QUESTION **What is geothermal energy, and how do people use it?**

KEY WORDS

geothermal energy — energy from _____ that is transferred from inside Earth.

TWO WAYS TO USE IT

WAY	WHAT IT DOES
heat homes and buildings	_____
make electricity (steam spins a turbine)	_____

ELECTRICITY PATHWAY

HEAT FROM EARTH

↓

BOILS WATER INTO STEAM

↓

SPINS A TURBINE

↓

TURNS A GENERATOR

↓

ELECTRICITY

CHECK YOUR THINKING

Where does geothermal energy come from?

Benefits and Limitations

Name: _____ **Date:** _____

BENEFITS AND LIMITATIONS

BENEFIT	LIMITATION
little pollution (clean)	_____
heat is always available underground	_____
needs a special location (near hot rock)	_____
costs a lot to build	_____

LOCATION MATTERS

Most steam and hot-water sources are near areas where Earth's crust is active, such as volcanic areas.

Geothermal works best in places with _____ activity.

CHECK YOUR THINKING

Why can't every place use geothermal energy?

CONNECT *Chapter 8 connection: Earth's own heat can warm buildings and make electricity — but only in the right places.*